

EXPERIMENT 7

Aim: Write a Program to implement Sliding Window Protocol for Selective Repeat ARQ.

Algorithm

Step 1: Start.

Step 2: Read the total number of frames (**n**).

Step 3: Read the sender window size (**w**).

Step 4: Send all frames within the current sender window.

Step 5: Receive an ACK for each frame.

Step 6: If a frame is acknowledged, mark it as received.

Step 7: If a frame is lost or not acknowledged, retransmit **only that frame**.

Step 8: Slide the sender window forward only when the first frame in the current window has been acknowledged.

Step 9: Repeat until all frames are transmitted successfully.

Step 10: Stop.

```
#include <stdio.h>
```

```
int main()
{
    int n, w;
    int ack[100] = {0};
    int i = 0, j, status;

    // Step 2: Read total number of frames
    printf("Enter the total number of frames: ");
    scanf("%d", &n);

    // Step 3: Read sender window size
    printf("Enter the sender window size: ");
    scanf("%d", &w);

    // Step 4 onwards
    while (i < n)
    {
        printf("\nSender Window: ");

        // Display current sender window
        for (j = i; j < i + w && j < n; j++)
        {
            printf("%d ", j);
        }

        printf("\n");

        // Step 4: Send all frames in current window
```

```

for (j = i; j < i + w && j < n; j++)
{
    if (ack[j] == 0)
    {
        printf("Frame %d sent.\n", j);
    }
}

// Step 5: Receive ACK for each frame
for (j = i; j < i + w && j < n; j++)
{
    if (ack[j] == 0)
    {
        printf("Enter ACK for Frame %d (1 = ACK, 0 = Lost): ",
               j);
        scanf("%d", &status);

        // Step 6: If acknowledged
        if (status == 1)
        {
            ack[j] = 1;
            printf("Frame %d acknowledged.\n", j);
        }
        else
        {
            printf("Frame %d lost.\n", j);
        }
    }
}

// Step 7: Retransmit only lost frames after timeout
for (j = i; j < i + w && j < n; j++)
{
    if (ack[j] == 0)
    {
        printf("\nTimeout for Frame %d.\n", j);
        printf("Retransmitting only Frame %d...\n", j);

        printf("Enter ACK for retransmitted Frame %d (1 = ACK): ",
               j);
        scanf("%d", &status);

        if (status == 1)
        {
            ack[j] = 1;
            printf("Frame %d acknowledged.\n", j);
        }
    }
}

// Step 8: Slide the window
while (i < n && ack[i] == 1)
{
    i++;
}

```

```
    }  
  
    // Step 10: Stop  
    printf("\nAll frames are transmitted successfully.\n");  
  
    return 0;  
}
```

Output: